U.S. Environmental Protection Agency
Office of Research and Development
Center for Public Health and Environmental Assessment
Pacific Ecological Systems Division

Quality Assurance Project Plan:

Soil Health Practices and Groundwater Nitrate Contamination

QA Track Identification

L-PESD-0034218-QP-1-0

QA Category: B

ORD National Program/Research Area/Output ID/Product ID:
SSWR.405 Nutrients in a Changing Climate, Regional-ORD Applied Research Proposal funded in 2022 entitled "Soil Health Practices' Impact on Groundwater Nitrate Contamination"

Approvals

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Revision History with Brief Descriptions
r1 (date):
r3 (date):
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r4 (date):

A. RESEARCH MANAGEMENT

A1 Organization

This work is funded through a Regional-ORD Applied Research Proposal funded in 2022 entitled “Soil Health Practices' Impact on Groundwater Nitrate Contamination”.

This research falls under the StRAP 4 SSWR Nutrients research area, and an example partnership to implement a nutrient reduction strategy (SSWR Output 405.4 “Implementation of Nutrient Reduction Strategies”) to address a groundwater nitrate contamination issue. This work builds upon existing work in StRAP 3 under the Nutrients Outputs SSWR 6.1.3 and 6.3.2, transferring knowledge and ideas from research in Oregon to Kansas groundwater nitrate contamination and surface water nutrient contamination.

A2 Problem Definition/Background

Drinking water nitrate contamination is a significant economic and public health concern in many agricultural areas across the US, with few tested best practices for improving source water quality. This project seeks to establish a monitoring framework for quantifying the benefits of soil health practices for groundwater nitrate mitigation, by comparing among established fields varying in tilling, fertilizer and cover crop practices. Improved source-water protection options facilitated by this research would allow stakeholders to leverage funding for BMP implementation by providing the required ability to forecast water quality improvements from the BMPs. These improved forecasting capabilities would open a method of achieving safe drinking water nitrate concentrations through implementing surface-level BMPs rather than treating contaminated source waters. These efforts would provide a cost-effective and environmentally friendly avenue to safe drinking water that would benefit many small/rural communities throughout the United States.

The overarching goal of this project is to examine nutrient management to improve surface and groundwater quality. We will assess the effectiveness of nutrient management and soil health practices for reducing nitrate contamination of groundwater through a sol health testing study with multiple stakeholder involvement. The proposed work addresses several nationally important information gaps in approaches to connect land use, current farm Best Management Practices (BMPs) and groundwater contamination.

The objective of this ROAR project is to evaluate the effectiveness of current best management practices for reducing nitrate leaching in groundwater by collecting new data to help connect soil health measures to potential for nitrate leaching. To reduce loading of nitrogen to groundwater and surface waters, the approaches needed to partner with affected communities are integrated science, outreach and management efforts.

Nitrate contamination of groundwater sources of drinking water poses a public health risk to communities throughout the United States. Many communities that are affected by nitrate contamination in their drinking water sources rely on small/rural public water supplies (PWSs) that cannot afford to treat their source waters for nitrate. As a result, many small/rural communities are forced to consume water that exceeds the safe drinking water limit established in the Safe Drinking Water Act (SDWA) or outsource their drinking water supply. Either way, the effects of nitrate contamination are costly and pose serious health risks to many Americans.

Without the ability to treat nitrate contamination, the only option for mitigating nitrate contamination in drinking water for many small/rural communities is to implement Best Management Practices (BMPs) that prevent, or greatly reduce, nitrate percolation into groundwater. Though research regarding the relationship between surface-level activities (primarily agricultural activities) and source water nitrate concentrations has shown that surface level BMPs can mitigate significant amounts of nitrate leaching from farm fields (Delgado et al. 2001), the degree to which these BMPs have been shown to mitigate nitrate contamination in groundwater remains less clear (Wassenar et al. 2006; Stites and Kraft 2000; Exner et al. 2014). Recent work in Denmark measured declines in groundwater nitrate that resulted from a variety of sustainable N management practices such as nutrient plans, optimum fertilizer rates, site-specific groundwater protection zones, cover crops and buffer zones (Hansen et al. 2017). The success of these management practices is greatly impacted by the function potential linked to biogeochemical processes such as denitrification, which transforms nitrates to N_2, a non-reactive end product. Reducing agricultural surplus and improving nutrient use efficiency together allowed this reversal of groundwater nitrate upward trends (Hansen et al. 2012).

Based on the above issues related to nitrate contamination of groundwater, we have identified the following research questions: Will the implementation of soil health related BMPs such as cover crops and precision fertilizer application result in measurable reductions in soil extractable N and N surplus? Do these practices result in decreases in groundwater nitrate levels in areas where these practices have been implemented over the longer term?

To address these questions, we propose to examine the role that agricultural best management practices such as cover crops and conservation tillage collectively can reduce agricultural N surplus and soil extractable N after harvest in western Great Plains farm fields. We will also use existing state or local data on groundwater nitrate concentrations to examine groundwater nitrate trends in areas where these BMPs have been applied. These ROAR efforts will allow the state to track progress in groundwater nitrate contamination and to identify practices and approaches that can lead to groundwater quality improvements.

Anticipated Results and Regional, State, Tribal, and/or Community Impact:

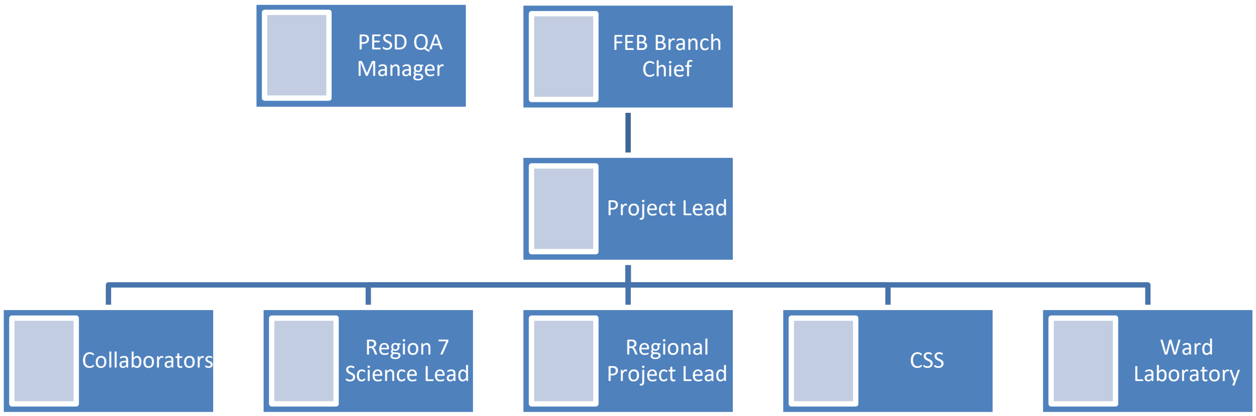
We anticipate that this research project will yield an accurate estimate of the degree to which no-till and cover crop BMPs mitigate nitrogen surpluses in fields which eventually lead to nitrate pollution in groundwater. Such quantification of potential groundwater quality benefits fostered by soil health BMPs will allow Source Water Protection (SWP) stakeholders to develop plans and leverage funding for BMP implementation. The ability to leverage such resources will help communities achieve safe nitrate concentrations in their sources of drinking water. This method of achieving safe drinking water nitrate concentrations through surface level BMP implementation will open a cost-effective and environmentally friendly avenue to safe drinking water that is not currently available to most small/rural communities throughout the United States.

Team Member Roles and Responsibilities:

<u>ORD project lead</u>: Will be responsible for addressing the overall goals through the research project design and implementation, will work with Regional Project lead to incorporate input from local partners and make decisions about the science, and will be responsible for leading the report-writing needed to share results to stakeholders. Will also be responsible for distributing, maintaining, and implementing this QAPP. The ORD Project lead is the TOCOR for the CSS contract task order(s).
<u>ORD collaborators</u>: Will be responsible for addressing the overall goals through the research project design and implementation (QAPP), and contribute to presentations and summary and report-writing needed to share results to stakeholders.
<u>RSL</u>: Ensure that the project is addressing the region's needs, meet with the project team at least quarterly for updates on progress.
<u>Regional project lead</u>: Help inform the overall goal(s) of the research. Recruit local/state partners to identify agricultural areas appropriate for this research. Oversee the local implementation of soil and well water sampling. Help draft methods for research design. Help analyze research findings and data. Help coordinate community presentations and distribute research findings to SWP stakeholders.
<u>QA Manager</u>: Responsible for reviewing and approving this QAPP. The QA Manager will not take part in any information collection or analysis activities and reports directly to the PESD

Division Director.

Figure 1. Project Organization Chart showing lines of communication and reporting relationships.



A3 Research Description

Research Objectives:

The goal of this research project is to provide methods to accurately calculate reductions in field-level N surplus, and soil nitrate concentrations after harvest through BMP implementation. These measures should quantify the potential mitigation of nitrate leaching that Source Water Protection (SWP) stakeholders can expect from implementing widely accepted soil health BMPs (like cover crops, no-till, or reduced nitrogen fertilizer application). By developing the methodology to accurately quantifying expected nitrate mitigation rates for one or two soil health BMPs, SWP stakeholders (in Region 7 and around the country) will have a tremendous asset in their SWP planning arsenal. Improved planning facilitated by this research would also allow SWP stakeholders to leverage funding for BMP implementation (i.e., through the Clean Water Act (CWA) section 319 grant program) that they would not have otherwise qualified for, due to a lack of water quality improvement forecasting abilities.

Research objectives are to:

1. Determine whether the implementation of soil health BMPs result in measurable reductions in soil nitrate test results and N surplus.
2. Determine whether reductions in soil nitrate concentrations and N surplus correlate with reductions in groundwater nitrate levels.
3. In areas with established soil health BMPs, assess trends in groundwater nitrate levels.

Research Approach:

The project will address the objectives above by measuring field-level soil extractable inorganic N (nitrate and ammonium) and N surplus as quick and easily measurable benchmarks, combined with nitrate measurements in groundwater from areas with long-term application of these BMPs. Through partnerships, we will identify sites where the cover crop practices have already been implemented so that we can look at the longer-term practice effectiveness and compare with historic groundwater nitrate data. We will then work with these partners to develop a framework for future groundwater monitoring to allow the state to track long-term responses to the future treatments. Additionally, some samples will be split and assessed using molecular methods to determine microbial activity linked to the nitrogen cycle. This information will be used to assess the potential impact of management practices on the functional genetic potential. The latter factors in the overall health and condition of soil, which are in turn influenced by interactions between members of the soil microbial communities in response to different levels of physico-chemical perturbations.

To study the degree of nitrate mitigation that one can expect from widely implementing soil health BMPs, we plan to calculate nitrogen budgets in fields with and without BMPs. The design will be somewhat flexible, depending on farmer partners, but we will have paired sites, with and without cover crops and precision fertilizer application. In one set of fields, agricultural producers will practice traditional tillage or fertilizer application, while in another set of fields, agricultural producers will not practice field tillage or will plant cover crops. Fields will have similar hydrologic functions and rainfall with only the management of the soil being different. By organizing this research project in the fashion outlined above, we hope to isolate the effect of cover crops to estimate field-level surplus and available nitrogen reductions associated with this practice.

ANTICIPATED/PLANNED ROAR PRODUCTS

Goal is to have data that will quantify the water quality improvements associated with BMPs in areas with nitrate concerns in Region 7.

Table 1. Anticipated/Planned ROAR Products

ROAR Product Type	ROAR Product Sub-Type	Product Description or Title	Due Date (Anticipated)	Final Product/ Deliverable? (Y/N)
Other	Fact sheet	Best management practices for nitrate mitigation and source water protection	One year after project begins, updated annually	N
Scientific Data, Software, and Models	Dataset	Collated data on cover crop impacts on soil health and potential impact on nitrate leaching	Two years after project begins	Y
Report	Summary	Analysis of existing soil health practices in Region 7 and their potential impact on nitrate leaching	Two years after project begins	Y
Presentations and Technical Summaries	Presentation to the region, state agencies and community	Best management practices for nitrate mitigation and source water protection	Two years after project begins	Y

A4 Quality Objectives and Criteria for Measurement Data

The measurement quality objectives for this project are to obtain valid data of documented and known quality to determine soil nutrient content, soil nutrient availability, soil carbon and nitrogen, soil carbon and nitrogen isotopes, phospholipid fatty acid concentrations (PLFA) and soil wet aggregate stability. All data will be carefully monitored for accuracy, precision, comparability, representativeness, and completeness.

Soil samples will be analyzed and reported following procedures outlined in the SOP for Soil Health Analysis (Appendix C) and the Ward Laboratory Qaulity Assurance Plan (Appendix B).

A5 Special Training Requirements/Certification

Personnel working on this project are required to have undergone basic field and lab safety training according to PESD policy. The Project Lead is responsible for assuring these trainings have been conducted. Training is provided in, and successful completion is documented by FedTalent. Basic knowledge of biogeochemistry, ecosystem ecology is helpful. All personnel working on this project are required to read and follow this QAPP and all SOPs. Personnel must be trained in the field and lab aspects of this work.

A6 Documentation and Records

This research falls under QA category B (research performed to solve specific problems or needs, i.e., applied research) and will include the record schedule: 1035b Environmental Programs and Projects and other record schedules as applicable (e.g., 1004 - Acquisitions).

Table 2. Records created during the project lifetime, record schedules and retention periods

<table>
<tr><td>Record Type</td><td>Record Schedule</td><td>Retention</td></tr>
<tr><td>Contract deliverables</td><td>1035a</td><td>Permanent</td></tr>
<tr><td>Research notebook</td><td rowspan="4">1035b</td><td rowspan="4">20 years from project completion</td></tr>
<tr><td>Data analysis files</td></tr>
<tr><td>Manuscripts (drafts, commented and final)</td></tr>
<tr><td>Laboratory QA Plan</td></tr>
<tr><td>QAPP</td><td rowspan="3">1035c</td><td rowspan="3">10 years from project completion</td></tr>
<tr><td>Laboratory SOPs</td></tr>
<tr><td>Sampling and analytical data files</td></tr>
<tr><td>Data and Field Sheets</td><td>1006g</td><td>After QC is complete</td></tr>
<tr><td>Email</td><td></td><td>10 years via Outlook</td></tr>
</table>

The Project Lead, Jana Compton, will be responsible for the project file. This QAPP will be reviewed annually for any dropped, added or altered activities by the project lead. If necessary, the QAPP will be revised then reviewed and reapproved by those who perform these activities initially. Any changes will be briefly described in the table immediately preceding Section A of the revised document.

The project file consists of the following: Documentation related to the Principal Lead's formulation and approval of the (research action plan, project and/or task), the selection of the research methodology, quality assurance task plans, raw data, laboratory notebook, any project- or study-related correspondence, or other data collection media, copies of interim reports showing data tabulation results and interpretations, copies of the final reports, peer reviews, and quality assurance assessments.

B. MEASUREMENT / DATA ACQUISITION

All soil samples of will be taken from previously agreed upon sites with verbal permission of the landowner through KDHE and EPA Region 7.

B1 Sampling Process Design (Experimental Design)

All samples of soil will be taken from previously agreed upon fields with the permission of the landowner. All sites were chosen based on the cooperation of the landowner and to obtain a mix of sites with and without cover crops. There may be a range in ages of the cover crop duration as well. The field plan requires that the landowners will provide some information about the crop management and site history (see below).

Soil sampling will occur once after harvest of the cash crop, in mid-November 2023. Then the sampling will be repeated during the early growing season in 2024 (pre-fertilizer, dates to be determined and will vary by crop type). We will used a grid-based sampling approach.

The study will also request participating farmers to fill out a form gathering field practice information to allow calculation of a nitrogen budget for participating fields, and accurately interpret the soil test.

Plot layout and locations

There are three properties containing fields; W, E, and S. W has 1 field, E has 3, and S has 4. For a total of 8 fields. Maps will be printed for every sampling group along with a list of corresponding coordinates. Within each field, 6-10 plots will be outlined, collecting 9 samples within each plot that will be composited (Figure 1), and this will be repeated 10 times in 8 fields across the ~20 acre farm field to generate 10 composite soil samples per field (Figure 1).

This brings us to a total of three Properties (W,E, and S), containing a total of 68 plots which will contain 612 cores. Naming convention would be Property (W, E, or S) followed immediately by Field Number (1-4), and finally by plot letter (A-J)

Table 3. List of fields, plots and subplots that will be sampled during the ROAR project.

Field Codes	Plot information	Number of subplots
W1	W1—includes 8 Plots (A-H)	9 subplots
E1	E1- includes 9 plots (A-I)	9 subplots
E2	E2- includes 7 plots (A-G)	9 subplots
E3	E3- includes 9 plots (A-I)	9 subplots
S1	S1- includes 10 plots (A-J)	9 subplots
S2	S2- includes 10 plots (A-J)	9 subplots
S3	S3- due to it's odd shape includes 6 plots (A-F)	9 subplots
S4	S4-include 10 plots (A-J)	9 subplots

Ex. W1_C as we will be compositing the subplots there's no need to identify them, however the coordinates will be provided for each subplot centroid and the identification will be:

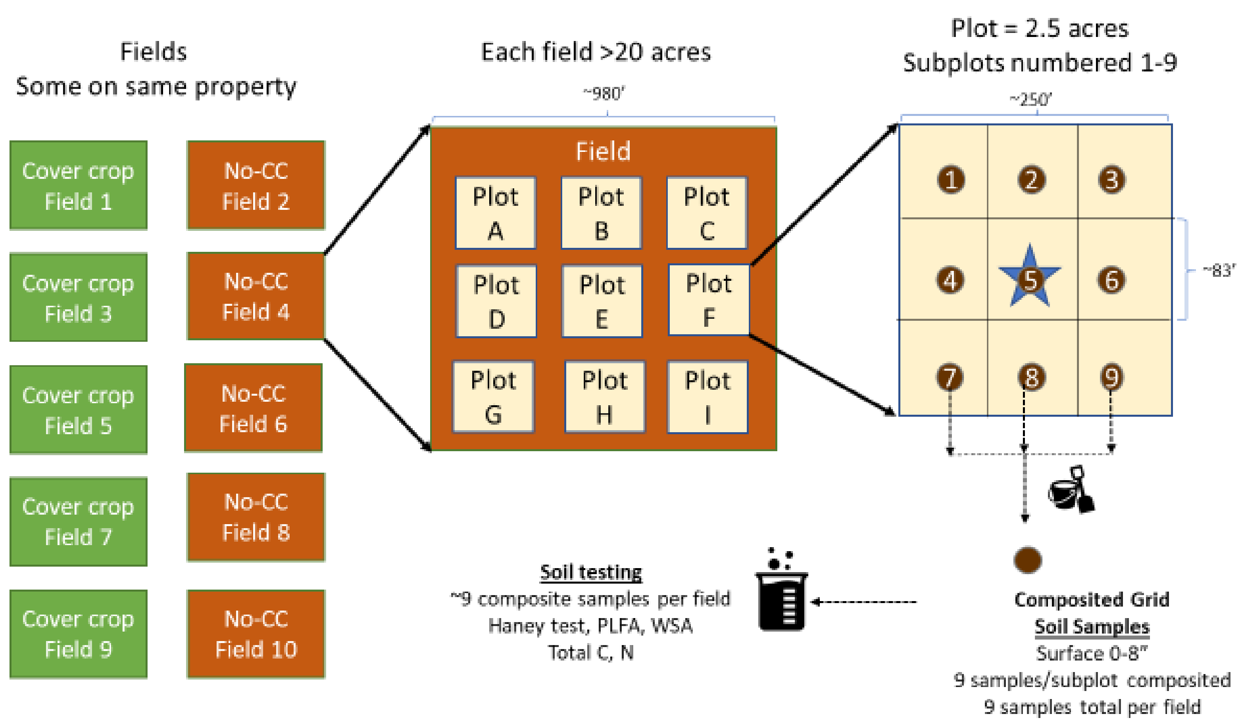
Subplot ex:

a	b	c
d	e	f
g	h	i

Crews will use the coordinates for subplot "e" to begin the sampling, then measure using a range finder or tape to find the next location.

There are a few instances in which the total distance between subplots may put crews in a waterway or just off the field edge. In those instances, the crews will still sample, but from the nearest planted area to the intended location.

Figure 2. Soil sampling layout as proposed. The blue star is the geolocated plot center.



Samples collected will be 0-8 inches of mineral soil. Soil sampling will be conducted using a 1-inch diameter stainless steel punch probe from 0-8 inch depth, 9-10 locations per site (locations to be provided by EPA Region 7). The number of plots or subplots may vary depending on the field configurations on the landscape; size of subplots will be kept constant.

The soil core samples will be placed in a zipperlock bag, stored on ice, and delivered to the lab for the following analyses:

- Haney Soil test to determine available soil nutrients for plants and microbes (Ward Laboratory)
- Phospholipid Fatty Acid Test (PLFA) soil microbial community structure and abundance (Ward Laboratory)
- Wet Aggregate Stability test (Ward Laboratory)
- Total Soil Carbon and Nitrogen (at EPA ISIRF lab)

These sites will break down to a total of 3 landowners and approximately 10 sites. The cash crop is primarily corn grown for grain and the cover crop is to be determined.

The EPA ISIRF laboratory operates under a Quality Assurance Manual (L-PESD-0032949-QM-1-1), SOPs (see Table 5) and participates in yearly performance testing programs. Ward Laboratory was recommended by Kansas Department of Health and the Environment partners, and Ward provided an unsigned version of their Quality Assurance Manual. A copy of the approval page with signatures was requested and if received, will be added as an appendix to this QAPP. Ward is accredited by the State of Nebraska and participates in at least two soil proficiency testing programs.

<u>The following information will be obtained from participating farmers:</u>

- Field name (no farmer names or addresses to protect CUI).
- Allow contractor to sample soil in participating fields fall 2023, and spring 2024.
- Crop History
- Dried sample of corn grain from the participating fields, near harvest time
- Field fertilization history (timing and amount, 2022-2023)
- If applicable, years using cover crop in participating fields, cover crop mix, and termination method.
- Cash crop rotation schedule and planting methods.
- 2023 Harvest details and yield: Crop variety, and biomass of crop harvest.
- Plant samples for nitrogen content of harvested crop removed from the field.
- Well water sample (optional if you would like nitrate analysis on your well water).

A data sheet will be prepared for the farmers to fill out about their land. These data will be obtained by Kansas Department of Health and the Environment, and Region 7, and provided without farmer names, addresses or any other identifying information aside from the field numbering.

B2 Sampling Methods Requirements

The sampling process is described below.

Wear work gloves to conduct the labor, and bring rain gear and comfortable, warm and potentially waterproof footwear depending on conditions. Layers will be ideal for spring and winter conditions in Kansas.

Soil Sampling Procedure

1. To take a soil sample, you will need a soil punch probe and a clean plastic bucket. Also have zipperlock bags (quart of gallon sized). Pre-label the sampling depth on the corer using a sharpie or tape. Check the depth label each day to ensure it is legible.

2. Use a large knife or trowel to scrape away the O horizon after clearing litter prior to sampling.

3. Collect 9 punch samples, 0 to 8 inches deep from each plot as shown in Figure 1.

4. If cropping, fertilizing, and/or liming has not been applied uniformly in a field, then a separate sample should be taken from each management or soil area. If soil areas within a field are different in appearance (slope, drainage, color, or texture) each area should be sampled separately. Small areas may not need to be sampled, but they will give some indication of the variation within the field.

5. Mix the cores thoroughly in sample bag. Label the bags using the Site Codes from Table 3 and use this same identification on the soil information sheet (i.e. chain of custody, see Appendix A).

6. Sample separately to avoid such areas as dead furrows, alkali spots and terrace channels.

7. Gently squeeze excess air out of the bag while sealing to minimize heterotrophic activity in the sample during transport, and to save room in the coolers. Ensure label is accurate and place in cooler with ice. If using ice, physically separate the ice from the sample bags by double bagging to avoid the bags becoming waterlogged.

8. Wipe or brush off soil probe between subplot samples. Use soap and water to clean the probe and bucket. Dry the probe and bucket prior to moving to a new plot.

9. If the soil information sheet was filled out as a hard copy, place it in a zipperlock bag and tape it inside the lid of the cooler. If the soil information sheet was filled out electronically, email the sheet to Jordan Westengaard at jwestengaard@wardlab.com and cc Jana Compton at Compton.jana@epa.gov.

10. Seal the cooler with tape so it cannot accidentally open during transport. Seal the cooler drain. Ship samples via UPS or FedEX to: Ward Laboratories, 4007 Cherry Ave, Kearney, Nebraska 68847, and maintain the shipping receipt.

B3 Sample Handling and Custody Requirements

Ward Laboratory sampling handling will follow their Quality Assurance Manual Section 4.5. ISIRF sample handling will follow their SOP.

Table 4 is a summary of sampling containers, preservation requirements and holding times. The sample collections will be in accordance with the attached standard operating procedures as listed in Table 5.

Table 4. Summary of sample collection, handling, and preservation activities

Sample Type	Parameters Measured	Sample Container	Minimum Sample Size	Preservation Method/Storag	Maximum Holding Time
Soil	Haney Test, Soil respiration, PLFA, Wet aggregate stability	Zipperlock bag	100 g	Store on ice until shipping Air drying in lab	Indefinite for air-dried samples

B4 Analytical Methods Requirements (including statistics if appropriate)

Summary of the requested analytical parameters and methods is provided in Tables 4 and 5. Standard laboratory operating procedures will be followed during the analysis of the samples at Ward lab, including QA/QC measures and equipment inspection/maintenance.

Table 5. Analytical parameters and Standard Operating Procedures (SOPs) used in this project.

Agency of Origin	QA Document Number/version	Title	Approval Date	Last Review Date
Ward Labs	SOP 1114	Haney Soil Health Procedure (H2O & H3A Extraction)	Oct 20, 2014	Dec 17, 2019
Ward Labs	SOP 1115	Soil respiration test	Jan 20, 2018	Sept 1, 2023
Ward Labs	SOP 1130	Buyer-Sasser Fatty Acid Methyl Ester (FAME), Phospholipid Fatty Acids (PLFA) and Neutral-Lipid Fatty Acid (NLFA) test	Dec 15, 2021	Dec 15, 2021
Ward Labs	SOP 1121	Wet Aggregate Stability test	March 21, 2019	Jan 21, 2020
US EPA ISIRF lab	PESD-EEB-SOP-4420-0	Isotopic analyses of environmental samples for $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ by EA-IRMS	April 18, 2022	April 18, 2022

B5 Quality Control Requirements

See Ward Laboratory Quality Assurance Manual Section 8.4 (Appendix B) for details.

B6 Instrument/Equip. Testing, Inspection, Maintenance Requirements

See Ward Laboratory Quality Assurance Manual Section 5.2 for details.

B7 Instrument Calibration and Frequency

See Ward Laboratory Quality Assurance Manual Section 6 for details.

B8 Inspection/Accept. Requirements for Supplies and Consumables

See Ward Laboratory Quality Assurance Manual Section 6.3 for details.

B9 Data Acquisition Requirements (Non-direct Measurements)

The Oregon DEQ has long term groundwater nitrate data that will be analyzed as part of this project. As collection, analysis and reporting followed their SOPs, the data is of sufficient quality for use in this project.

B10 Data Management

Field data will be recorded on field sheets by the CSS field staff. Every day after returning to the office, newly transcribed data will be scanned as a backup copy and saved in a folder titled with that sampling date and to OneNote. Data from field notebooks will also be transcribed into Microsoft Excel spreadsheets. All data and field sheets will be uploaded into the OneNote identified in Section A6.

Samples will be taken, packaged and shipped according to protocol and budget allowance to analytical laboratories. Analytical data generated by each laboratory will be sent to the project managers in a format appropriate (pdfs for field data sheets, analytical data will be provided in Excel files). Data will be analyzed and formatted for project appropriate using Excel.

All data related to the Soil Health ROAR project will be kept on the following shared folder at EPA- PESD:
L:\Priv\CORFiles\Projects\ROAR KS Soil Health
Access is restricted only to EPA staff working on this project.

Project notes and information are also shared via a MS Teams site to facilitate sharing with state partners, but our final information repository will be the EPA network shared folder shown above.

Landowner name and address will be kept separate from any data and confidential so that it is not associated with data. For example, names and addresses will not be included in naming conventions for fields, nor placed on the data sheets or files. Landowner name and address will be kept confidential using the CUI marking in Word to the extent permitted by law, and kept in password protected files separate from the data.

Soil data collected from the farms would be shared with the participating farmer, and any identifying information about field location or identification will be kept confidential.

C. ASSESSMENT / OVERSIGHT

C1 Assessments and Response Actions

The PI will conduct annual reviews and assessments of the progress and accomplishments of all aspects of the research. This review will be the basis for making decisions on research activities for the upcoming year. The research is subject to periodic project level QA assessments by the EPA QAM. Reports of the assessments will be provided to the Project Lead and any needed corrective actions will be discussed and agreed upon between the project Lead And QAM.

Overall project assessment and oversight, including field activities will be the responsibility of the project manager. Laboratory assessment and oversight will be provided by Quality Assurance Managers at EPA and Ward Laboratories. The project manager will be notified of any delays, analytical anomalies or data quality limitations.

C2 Reports to Management

The PI reports to the RAP Project and/or Product Leader and to the Branch Chief. In addition, the ROAR project has numerous reporting requirements to the Regional Science Liaison in Region 7 and the Regional Program contact.

Table 6. Gantt chart for tracking progress of the overall ROAR project.

Task	2022	2023				2024				2025
	Fall	Winter	Spring	Summer	Fall	Winter	Spring	Summer	Fall	Winter
Identify partners, Establish contracts										
Site selection										
Soil sampling										
Soil Health Testing										
Soil C and N Testing										
Data analysis										
Writing reports										
Final reports and presentations										

D. DATA VALIDATION AND USABILITY

D1/D2 Data Review, Validation, and Verification Requirements and Methods

ISIRF and Ward laboratory data review, verification, and reporting will follow their Quality Assurance Plans. Both labs will provide data in electronic format with quality assurance information, sample receiving information and a project narrative. Any data that may be inaccurate, misleading, or otherwise fails the laboratory's quality standards due to field or sampling activities will be identified in the final analytical report.

The EPA will provide standard data review, verification and validation on all analytical data generated by this project. The task lead will review all data received from the laboratories for completeness, quality control outliers and results outside of historical bounds by scanning the Excel spreadsheets and plotting the results. The task lead may ask the laboratory to investigate results outside of historical bounds to determine if there was a sample misidentification or error in the analysis/reporting process.

D3 Reconciliation with User Requirements

Data collected in this project will evaluate the effectiveness of current BMPs in reducing nitrate leaching. Data not meeting the Ward Laboratory quality control criteria will be assessed for its affect (percent bias or deviation) on our evaluation. The results may still be used if deemed needed for final analysis. These decisions will be made on a case-by-case basis and will be documented in the data spreadsheet.

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Appendix A - Ward Lab Electronic Chain of Custody

Soil_Health_sample_s
ubmission_fillable_05

Appendix B - Ward Lab Quality Assurance Manual

SOP_3000
Quality_Assurance_Pl

Appendix C – Ward Laboratory Soil Health Lab Testing Protocols

SOP_1114 Haney Soil Health Procedure

SOP_1115 Soil Respiration Test.pdf

SOP_1121 Wet Aggregate Stability.pc

SOP_1130 PLFA NLFA FAME.pdf